- How many Ruby on Rails application instances are running? How many threads are configured per process? Each thread acquires a database connection to complete work.
- What is the connection pool size per process? Multiply the number of processes by the configured pool size.
- How many Sidekiq processes are running? How many threads are running per process? Sum the total number of threads to get the total concurrent number of transactions that are possible.

Complete this type of capacity planning analysis, totaling up web processes, background processes, and looking at their pool sizes, to gain a possible maximum theoretical number of concurrent connections the application would use.

Compare that figure with how you've configured max_connections. Under provision, leaving some connections available for administrative work like a psql session or other PostgreSQL CLI programs like vacuumdb. This also assumes there are not other client applications connected to the same PostgreSQL database. As a best practice, give each client application a unique database.

When you're running out of connections, you'll need more help. Fortunately there is additional software you can run for such a scenario.

Working With PgBouncer

When the concurrency of your application grows, the number of physical database connections used by the Active Record Connection Pool alone may not be enough.

PgBouncer, free open source software that you can run in between Ruby on Rails and PostgreSQL, allows you to scale to higher levels of concurrent database activity.

- Client connections may exceed max_connections
- Connections can be held open even if the database server is temporarily unreachable, delaying their processing but not refusing the connections
- The pooler re-uses connections more efficiently, achieving higher levels of concurrency on the same compute resources
- Connection Poolers may keep connections open while backends are modified, for example when modifying the server instance

In this section you'll install, configure, and begin using PgBouncer with Rideshare.